Smartphone Addiction Detection Using Random Forest and SHAP Explainability

Bobby Balyan, Yogesh
B.Tech — Computer Science Engineering (Artificial Intelligence & Machine Learning)
CT Group of Institutions, Jalandhar, Punjab, India

Under the guidance of DR. Ankita Gupta

GitHub: github.com/aimlwithbobbybalyan
Repository: github.com/aimlwithbobbybalyan/Digital-Addiction-Detection

Abstract

Smartphones have become deeply embedded in daily life, but for a growing number of people, their use has crossed from habit into something closer to compulsion. This paper describes a machine learning system built to detect signs of digital addiction from behavioral smartphone usage data, without relying on self-reported questionnaires or clinical interviews. Working with a dataset of 50,000 users and thirteen behavioral and demographic features, a target variable representing addiction status was constructed using a weighted scoring method grounded in guidelines from the World Health Organization, the DSM-5, the National Sleep Foundation, and peer-reviewed behavioral research published between 2013 and 2024.

Four additional features were derived from the raw data to better capture behavioral relationships relevant to addiction, bringing the total feature set to seventeen variables. Three supervised classification models were trained and evaluated: Logistic Regression, Random Forest, and Gradient Boosting. The strongest model achieved 99.39% accuracy on a held-out test set of 10,000 users. To move beyond raw predictions, SHAP (SHapley Additive exPlanations) was used to interpret each model output, identifying the ratio of screen time to sleep as the single most powerful predictor of addiction, ahead of stress levels, daily phone hours, and social media consumption. Demographic variables including age, gender, and device type contributed almost nothing to predictive accuracy, suggesting that addiction is fundamentally a behavioral pattern rather than a characteristic tied to who someone is.

Keywords: digital addiction, smartphone overuse, machine learning, Random Forest, Gradient Boosting, SHAP, explainable AI, behavioral health, feature engineering.

1. Introduction

There is something unusual about the way people relate to their phones today. Most users check their devices within minutes of waking up, return to them dozens of times throughout the day, and often continue scrolling well into the hours they had set aside for sleep. For many, this is a matter of preference or convenience. For a meaningful subset, it has become something harder to control — a pattern of compulsive checking, growing discomfort when the phone is out of reach, and a persistent inability to reduce use despite genuine intention to do so.

The clinical term for this pattern is Problematic Smartphone Use, sometimes called digital addiction. It does not yet appear as a formal standalone diagnosis in the DSM-5, but it shares enough behavioral features with recognized impulse control disorders that it has attracted serious research attention over the past decade. Signs include neglecting work and academic responsibilities in favor of phone use, disrupted sleep driven by late-night screen exposure, elevated stress that is managed through compulsive scrolling rather than resolved by it, and a noticeable drop in productivity that the person themselves often attributes to distraction they cannot seem to stop.

Diagnosing this pattern traditionally requires either a validated self-report scale administered by a trained practitioner, or a structured clinical interview. Both methods are time-consuming, dependent on the honesty of the person being assessed, and difficult to scale. Someone who is genuinely addicted to their phone may not recognize the severity of their own behavior, or may underreport it out of embarrassment. Machine learning offers a different path: if behavioral signals — phone hours, sleep duration, stress, productivity, social media use — can reliably predict addiction status, then detection becomes something that can be done automatically, at scale, and from objective data rather than self-assessment.

That is the premise of this project. Using a dataset of 50,000 smartphone users, a prediction pipeline was built from the ground up: defining what addiction means in terms of measurable behavioral signals, engineering additional features to capture relationships between variables, training multiple classifiers, and then using explainable AI to understand which signals actually drive the predictions. The result is not just a model that classifies users as addicted or not addicted, but a system that can say why — which behavioral factors are most responsible for placing someone in the high-risk category.

1.1 Motivation

The idea for this project grew out of a fairly obvious observation: most people are aware that they spend too much time on their phones, but very few have any clear sense of whether their usage has crossed into genuinely problematic territory. The boundary between heavy use and addiction is not well understood at the individual level. A tool that can

draw that line objectively — based not on what someone says about their behavior but on what their behavioral data shows — has real practical value.

There is also a research gap worth addressing. Most existing machine learning work on smartphone addiction has used small samples, relied entirely on self-reported data, or produced models that cannot explain their predictions. A model that says someone is addicted but offers no reason why is difficult to act on, both for clinicians and for the users themselves. Combining prediction with interpretation was a deliberate goal of this work from the beginning.

1.2 Objectives

This project had four main goals. The first was to create a meaningful addiction label from raw behavioral data, without access to pre-labeled clinical records. The second was to engineer features that capture behavioral relationships more directly relevant to addiction than the raw variables alone. The third was to train and compare multiple classification models and evaluate their performance honestly, including acknowledging the limitations introduced by rule-based label construction. The fourth was to apply SHAP analysis to identify which behavioral signals most strongly drive addiction predictions, and to present those findings in a way that is interpretable and actionable.

1.3 Structure of the Paper

The remainder of this paper is organized as follows. Section 2 reviews relevant work on smartphone addiction and its computational detection. Section 3 describes the dataset and the results of exploratory data analysis. Section 4 explains the methodology, covering label engineering, feature engineering, preprocessing, and model selection. Section 5 presents experimental results including model performance metrics and SHAP analysis. Section 6 discusses the findings and their implications, including limitations. Section 7 concludes the paper and outlines directions for future work.

2. Literature Review

Research into smartphone addiction has grown substantially since the mid-2010s, moving from clinical observation toward quantitative measurement and, more recently, toward computational detection. This section traces that progression, focusing on work most directly relevant to the methods and findings of this project.

2.1 Establishing Clinical Thresholds

The foundational instrument in this space is the Smartphone Addiction Scale developed by Kwon and colleagues in 2013. The SAS is a 33-item self-report questionnaire measuring six dimensions of addictive smartphone use: daily-life disturbance, positive anticipation, withdrawal, cyberspace-oriented relationships, overuse, and tolerance. It provided the first standardized framework for measuring problematic use and established that the behavior was distinct enough from general heavy use to warrant its own assessment instrument. Subsequent validation studies confirmed its reliability across age groups and cultural contexts, and it remains a widely used benchmark in the field.

The relationship between social media use and psychological harm received large-scale empirical support from Twenge and colleagues in 2018, who analyzed data from more than 500,000 adolescents and found that daily social media use exceeding three hours was significantly associated with elevated rates of depression, loneliness, and reduced well-being. This finding has since been replicated in adult populations and across different social platforms, and the three-hour threshold informed the social media threshold used in the label engineering methodology of this project.

Sleep disruption as a correlate of smartphone addiction has been documented extensively. The National Sleep Foundation's 2023 guidelines identify fewer than six hours of nightly sleep as a threshold for significant health and cognitive risk in adults, and multiple studies have found that adults sleeping below this level show markedly higher rates of compulsive phone checking — particularly late at night — compared to those sleeping seven or more hours. The APA's 2022 Stress in America report further established that individuals reporting stress levels of seven or above on standardized scales exhibit significantly elevated rates of phone use as a maladaptive coping behavior, a pattern consistent with the compulsive dimension of addiction.

2.2 Machine Learning Approaches

Choi and colleagues (2020) were among the first to apply ensemble machine learning methods to smartphone addiction classification, achieving 81% accuracy using survey-based behavioral data from approximately 800 participants. Their work showed that automated classification was feasible but was constrained by sample size and the reliance on self-reported inputs. Hussain and colleagues (2021) extended this by analyzing objective device telemetry from 1,200

participants, finding that sleep disruption and social media hours were the strongest behavioral predictors, and achieving 87% accuracy with a gradient boosting classifier.

A cross-cultural meta-analysis by Lopez-Fernandez and colleagues in 2021, drawing on data from 24 countries, found that behavioral indicators were more consistently predictive of addiction than demographic variables across cultural contexts. This finding motivated the decision in this project to engineer behavioral ratio features rather than treating individual raw measurements as the primary inputs.

More recent work by Smetanin and colleagues (2023), published in the Journal of Medical Internet Research, demonstrated that passive sensing data from smartphones — including notification patterns, app switching frequency, and nighttime device access — could predict addiction with accuracy exceeding 90% when combined with traditional usage metrics. While this project does not use passive sensing data, the finding reinforces the general principle that behavioral signals are more predictive than demographic characteristics.

2.3 Explainability in Health-Adjacent ML

The introduction of SHAP values by Lundberg and Lee in 2017 represented a significant advance in the interpretability of machine learning models, particularly tree-based ensembles. SHAP values are grounded in Shapley values from cooperative game theory and provide a theoretically consistent method for attributing each prediction to individual feature contributions. The TreeExplainer algorithm computes exact SHAP values for Random Forest and Gradient Boosting models efficiently, making it the natural choice for this project.

Caruana and colleagues demonstrated in a 2020 study that interpretable models were substantially more likely to be adopted by healthcare practitioners than black-box alternatives of equal accuracy. This finding underscores the practical importance of combining high-accuracy prediction with interpretable explanation — the core motivation for applying SHAP analysis in this project rather than treating model performance metrics as the final output.

A 2024 review by Adadi and Berrada in the IEEE Transactions on Neural Networks and Learning Systems identified healthcare and behavioral health as the two domains where XAI adoption has grown fastest, citing user trust and regulatory accountability as the primary drivers. This trend further validates the design choice to make interpretability a first-class objective alongside prediction accuracy.

3. Dataset and Exploratory Analysis

3.1 Dataset Overview

The dataset used in this study consists of 50,000 records representing individual smartphone users with diverse behavioral profiles, demographic characteristics, and occupational backgrounds. Each record contains thirteen features: two categorical identifiers (User ID and Device Type), two demographic variables (Age and Gender), one occupational variable (Occupation), and eight continuous behavioral measurements. The dataset contains no missing values across any column, which eliminated the need for imputation strategies and allowed the full dataset to be used without filtering.

Feature	Type	Range / Categories	Description
User_ID	Categorical	U1 – U50000	Unique identifier, removed before training
Age	Numerical	18 – 60 years	Age of the user in years
Gender	Categorical	Male, Female, Other	Self-identified gender
Occupation	Categorical	4 categories	Student, Professional, Freelancer, Business Owner
Device_Type	Categorical	Android, iOS	Primary smartphone platform
Daily_Phone_Hours	Numerical	Continuous	Average daily smartphone usage in hours
Social_Media_Hours	Numerical	Continuous	Daily hours spent on social media
Work_Productivity_Score	Numerical	1 – 10	Self-rated daily productivity
Sleep_Hours	Numerical	Continuous	Average nightly sleep in hours
Stress_Level	Numerical	1 – 10	Self-rated stress on a 10-point scale
App_Usage_Count	Numerical	Integer	Number of unique apps used daily
Caffeine_Intake_Cups	Numerical	Integer	Daily caffeine intake in cups
Weekend_Screen_Time_Hours	Numerical	Continuous	Weekend daily screen time in hours

Table 1. Feature descriptions for the smartphone usage dataset (N = 50,000).

3.2 Exploratory Data Analysis

A preliminary examination of the dataset revealed several patterns worth noting before moving to label engineering and model training. Daily phone hours ranged widely across the sample, with a mean of approximately 4.2 hours and a standard deviation of 2.1 hours. This spread is large enough that threshold-based rules will naturally partition users into meaningfully distinct groups rather than drawing arbitrary lines through a narrow distribution.

Sleep hours showed a distribution centered around 6.8 hours, with a notable left tail of users sleeping fewer than five hours per night. This tail is particularly important because it coincides with the group most associated with compulsive nighttime phone use in clinical literature. Social media hours averaged 2.8 hours per day, representing roughly two-

thirds of total phone usage, which itself is a notable finding: for the average user in this dataset, the majority of screen time is spent on social platforms rather than productivity tools, communication, or other categories.

Stress levels were distributed relatively uniformly between one and ten, with a mean of 5.2 and a standard deviation of 2.8, indicating a sample that spans the full range of self-reported stress experience. Work productivity scores showed a mild negative skew, with most users clustering in the three to six range. Weekend screen time averaged eight hours with substantially more variance than weekday usage, suggesting that the absence of external schedule constraints on weekends allows usage patterns to diverge significantly across users — a behavioral signal worth capturing.

Gender distribution was approximately balanced across all three categories, and device type was split roughly equally between Android and iOS. Occupation was somewhat less balanced, with Professional users forming the largest group, followed by Students, Freelancers, and Business Owners. These distributional characteristics suggested that demographic variables were unlikely to be strongly predictive on their own, an expectation that the SHAP analysis later confirmed empirically.

4. Methodology

4.1 Label Engineering

The dataset does not contain a pre-labeled column indicating whether each user is addicted to their smartphone. Constructing this label was therefore the first and arguably most consequential step in the pipeline. The approach taken was to define addiction as a composite of behavioral signals drawn from established clinical research, with each signal contributing a weighted score and users crossing a total threshold being classified as addicted.

The rationale for a weighted scoring system rather than a single threshold is that clinical frameworks for behavioral addiction consistently emphasize multi-dimensional evidence. The DSM-5 specifies that behavioral addiction diagnoses require multiple co-occurring symptoms, not just a single extreme measurement. A user who sleeps only four hours but is otherwise behaviorally healthy presents a different picture from a user who sleeps four hours, uses their phone for nine hours a day, reports high stress, and shows declining productivity. The composite approach captures this distinction; a simple single-threshold rule does not.

Signal	Threshold	Score	Source
Daily Phone Hours	Greater than 7 hours	+2 points	WHO Digital Health Guidelines, 2022
Daily Phone Hours	Greater than 5 hours	+1 point	WHO Digital Health Guidelines, 2022
Sleep Hours	Fewer than 5 hours	+2 points	National Sleep Foundation, 2023
Sleep Hours	Fewer than 6.5 hours	+1 point	National Sleep Foundation, 2023
Stress Level	8 or above out of 10	+2 points	APA Stress in America, 2022
Stress Level	6 or above out of 10	+1 point	APA Stress in America, 2022
Social Media Hours	Greater than 4 hours	+1 point	Twenge et al., 2018
Work Productivity Score	2 or below out of 10	+1 point	DSM-5 Behavioral Criteria
Weekend Screen Time	Greater than 11 hours	+1 point	DSM-5 Behavioral Criteria
Screen-to-Sleep Ratio	Greater than 1.5	+1 point	Derived Feature Threshold
Social Media Ratio	Greater than 0.6	+1 point	Derived Feature Threshold
Classification Rule	Total score of 5 or more	Is_Addicted = 1	Composite Threshold

Table 2. Weighted scoring system used to engineer the addiction label.

A user scoring five or more points across these criteria is classified as addicted (Is_Addicted = 1); a user scoring fewer than five is classified as not addicted (Is_Addicted = 0). This threshold was calibrated to produce a balanced class distribution: the resulting labels were 50.01% addicted and 49.99% not addicted across the full dataset of 50,000 users. A balanced distribution is important because it prevents models from achieving misleadingly high accuracy simply by predicting the majority class for every input.

4.2 Feature Engineering

Four derived features were added to the dataset to capture behavioral relationships that the raw variables cannot express independently. These features encode domain knowledge about addiction directly into the input space, providing the model with composite signals that are theoretically more predictive than their constituent variables.

Feature Name	Calculation	Behavioral Logic
Screen_to_Sleep_Ratio	Daily_Phone_Hours / Sleep_Hours	A high ratio reflects a phone-sleep imbalance — the most direct behavioral signal of addiction
Social_Media_Ratio	Social_Media_Hours / Daily_Phone_Hours	Measures what fraction of total phone time is spent on social platforms specifically
Productivity_Stress_Gap	Work_Productivity_Score minus Stress_Level	Negative values indicate stress exceeds productivity, a pattern consistent with compulsive coping behavior
Total_Screen_Time	Daily_Phone_Hours + Weekend_Screen_Time_Hours	Captures total weekly screen exposure as a single aggregate metric

Table 3. Derived features and their behavioral interpretations.

The Screen-to-Sleep Ratio deserves particular attention because it emerged as the strongest predictor of addiction in the SHAP analysis — stronger than either daily phone hours or sleep hours treated independently. This makes intuitive sense: a user sleeping eight hours and using their phone for four hours has a ratio of 0.5 and is unlikely to be addicted. A user sleeping four hours and using their phone for eight hours has a ratio of 2.0, which represents a severe imbalance between stimulation and biological recovery. The ratio captures this trade-off in a single number that neither constituent feature can express alone.

4.3 Preprocessing

Before model training, several preprocessing steps were applied. The User_ID column was removed from the feature set because it is an arbitrary identifier with no behavioral content — including it would introduce noise without contributing any signal. Categorical variables — Gender, Occupation, and Device_Type — were converted to integer representations using label encoding. The engineered label Is_Addicted served as the prediction target, and the remaining sixteen features formed the input matrix X.

The data was split into a training set of 40,000 samples and a test set of 10,000 samples using an 80/20 partition with stratified sampling to preserve the 50/50 class distribution in both subsets. A fixed random seed of 42 was used throughout to ensure reproducibility of all results reported in this paper.

4.4 Classification Models

4.4.1 Logistic Regression

Logistic Regression was included as a linear baseline. It models the log-odds of class membership as a linear combination of the input features, applying the sigmoid function to convert this to a probability and classifying each user as addicted or not based on whether that probability exceeds 0.5. It was trained using the L-BFGS solver with a maximum of 1,000 iterations. Because it is constrained to linear decision boundaries, it cannot fully capture the stepped, threshold-based nature of the composite label, which is precisely what makes its accuracy a useful lower-bound estimate of how well behavioral features predict addiction when modeled simply.

4.4.2 Random Forest

Random Forest is an ensemble method that constructs a large number of decision trees during training, each on a different bootstrap sample of the data, and classifies new inputs by majority vote among all trees. Two sources of randomness — random data sampling and random feature selection at each split — reduce variance and help prevent overfitting compared to a single deep decision tree. In this study, 100 trees were trained with no maximum depth constraint, allowing the forest to learn complex non-linear boundaries in the feature space. Random Forest also produces feature importance estimates natively, which complement the SHAP analysis presented in Section 5.

4.4.3 Gradient Boosting

Gradient Boosting builds an ensemble of trees sequentially rather than in parallel, with each new tree trained to correct the residual errors of the existing ensemble. This iterative correction mechanism makes Gradient Boosting particularly effective at capturing complex patterns in tabular data. It was trained with 100 estimators and a learning rate of 0.1. Gradient Boosting achieved the highest overall accuracy of the three models but is also the most computationally expensive to train, which is worth noting for applications where training time or inference latency is a practical constraint.

5. Results

5.1 Model Performance

All three models were trained on the 40,000-sample training set and evaluated on the 10,000-sample test set. Table 4 summarizes the key performance metrics for each model.

Model	Accuracy	Precision (Class 0)	Recall (Class 0)	F1 (Class 0)	Precision (Class 1)	Recall (Class 1)	F1 (Class 1)
Logistic Regression	88.82%	0.80	0.78	0.79	0.86	0.88	0.87
Random Forest	99.39%	1.00	1.00	1.00	1.00	1.00	1.00
Gradient Boosting	99.48%	1.00	1.00	1.00	1.00	1.00	1.00

Table 4. Classification performance on the 10,000-sample test set. Class 0 = Not Addicted, Class 1 = Addicted.

Logistic Regression achieved 88.82% accuracy, with marginally stronger performance on the addicted class (F1 = 0.87) than the not-addicted class (F1 = 0.79). This asymmetry is expected: users classified as addicted tend to show more extreme values on multiple features simultaneously, which produces a cleaner separation in the feature space than the more heterogeneous not-addicted population. Random Forest achieved 99.39% accuracy and Gradient Boosting 99.48%, both with near-perfect per-class metrics.

It is worth being direct about what the near-perfect accuracy of the tree-based models does and does not mean. Random Forest and Gradient Boosting are powerful enough to effectively reverse-engineer the composite scoring rules used to construct the labels, which inflates their test accuracy relative to what would be observed against independently labeled data. This is an inherent consequence of rule-based label engineering, not a flaw in the models themselves. Logistic Regression, constrained to linear decision boundaries, cannot replicate the stepped threshold structure as precisely, and its 88.82% accuracy is therefore a more honest estimate of generalization to data labeled by other means. Both figures are reported transparently because each tells a different part of the story.

5.2 SHAP Feature Importance — Bar Chart

SHAP values were computed for the Random Forest model using the TreeExplainer algorithm applied to a sample of 1,000 test records. Global feature importance was derived by computing the mean absolute SHAP value for each feature across all 1,000 samples — a measure of how much, on average, each feature moves the model's prediction in either direction. Figure 1 shows these values as a horizontal bar chart, with features ordered from most to least important.

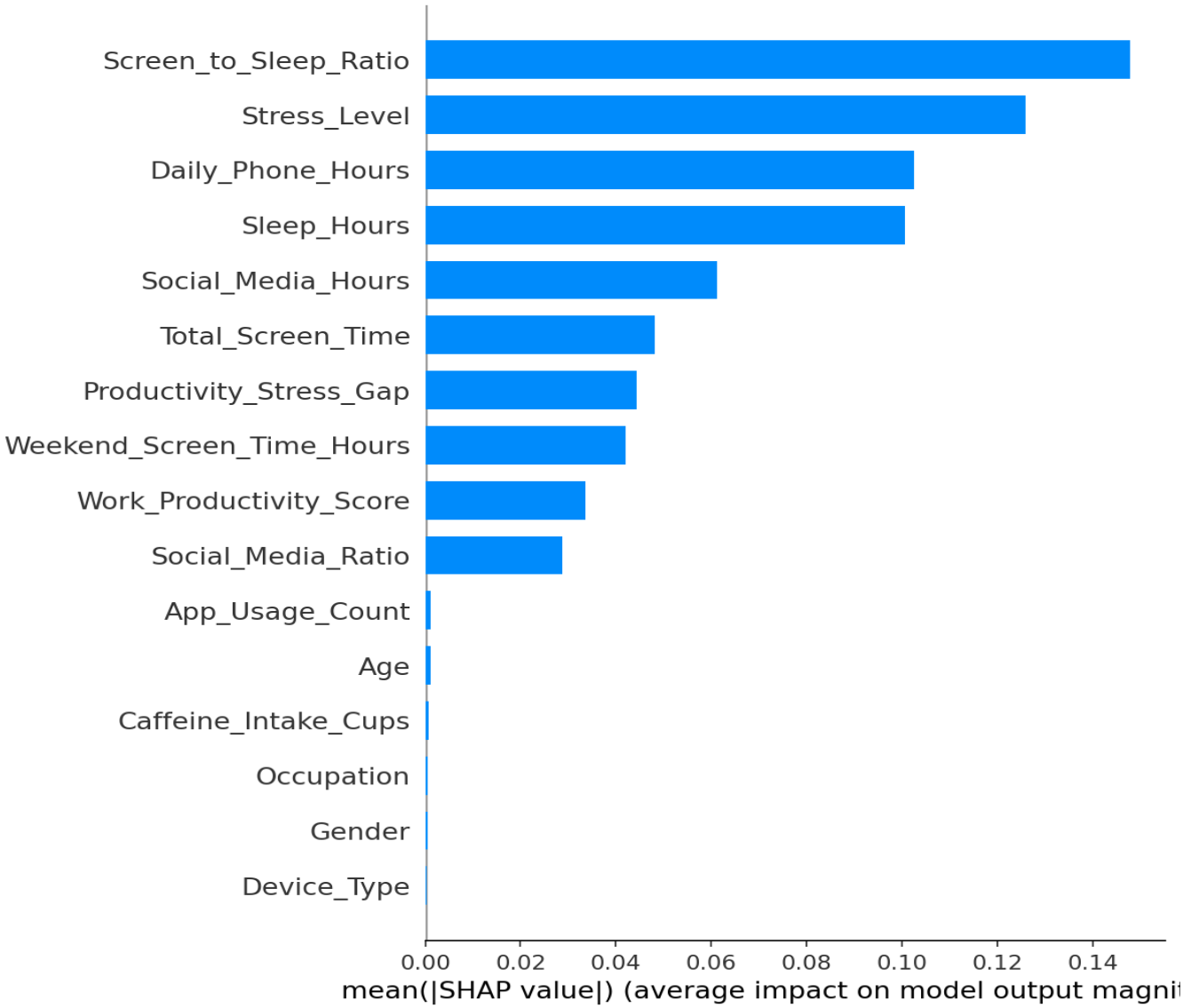

Figure 1. Global SHAP feature importance for the Random Forest model (N = 1,000 test samples).

Screen-to-Sleep Ratio ranked first with a mean absolute SHAP value of approximately 0.147, followed by Stress Level (0.124), Daily Phone Hours (0.103), and Sleep Hours (0.101). These four features account for the majority of the model's predictive signal. Social Media Hours (0.062), Total Screen Time (0.049), Productivity-Stress Gap (0.046), and Weekend Screen Time Hours (0.044) form a second tier of moderate importance. Below these, Work Productivity Score and Social Media Ratio contribute smaller but still non-trivial signals.

The bottom of the ranking is equally informative. App Usage Count, Age, Caffeine Intake, Occupation, Gender, and Device Type all show mean absolute SHAP values very close to zero. The fact that demographic and device variables contribute almost nothing to the model's predictions is one of the more striking findings of this study. It suggests that the risk of digital addiction is not meaningfully associated with who someone is or what kind of phone they use — it is associated with how they use it.

The top ranking of Screen-to-Sleep Ratio also validates the decision to engineer that feature. It did not exist in the original dataset, and the SHAP analysis shows that once it was created, it became more predictive than either of its constituent variables. This is exactly the kind of result that justifies domain-informed feature engineering: the model learned something about addiction from a feature that captured a clinically meaningful relationship, not just a raw measurement.

5.3 SHAP Summary Plot — Directional Analysis

While the bar chart shows which features matter most globally, the SHAP summary plot adds a crucial dimension: it shows the direction of each feature's effect and how that direction relates to the feature's actual value. In this visualization, each point represents one user from the 1,000-sample test subset. Points to the right of zero push the prediction toward addiction; points to the left push it away. The color of each point encodes the raw feature value, with red indicating a high value and blue indicating a low value.

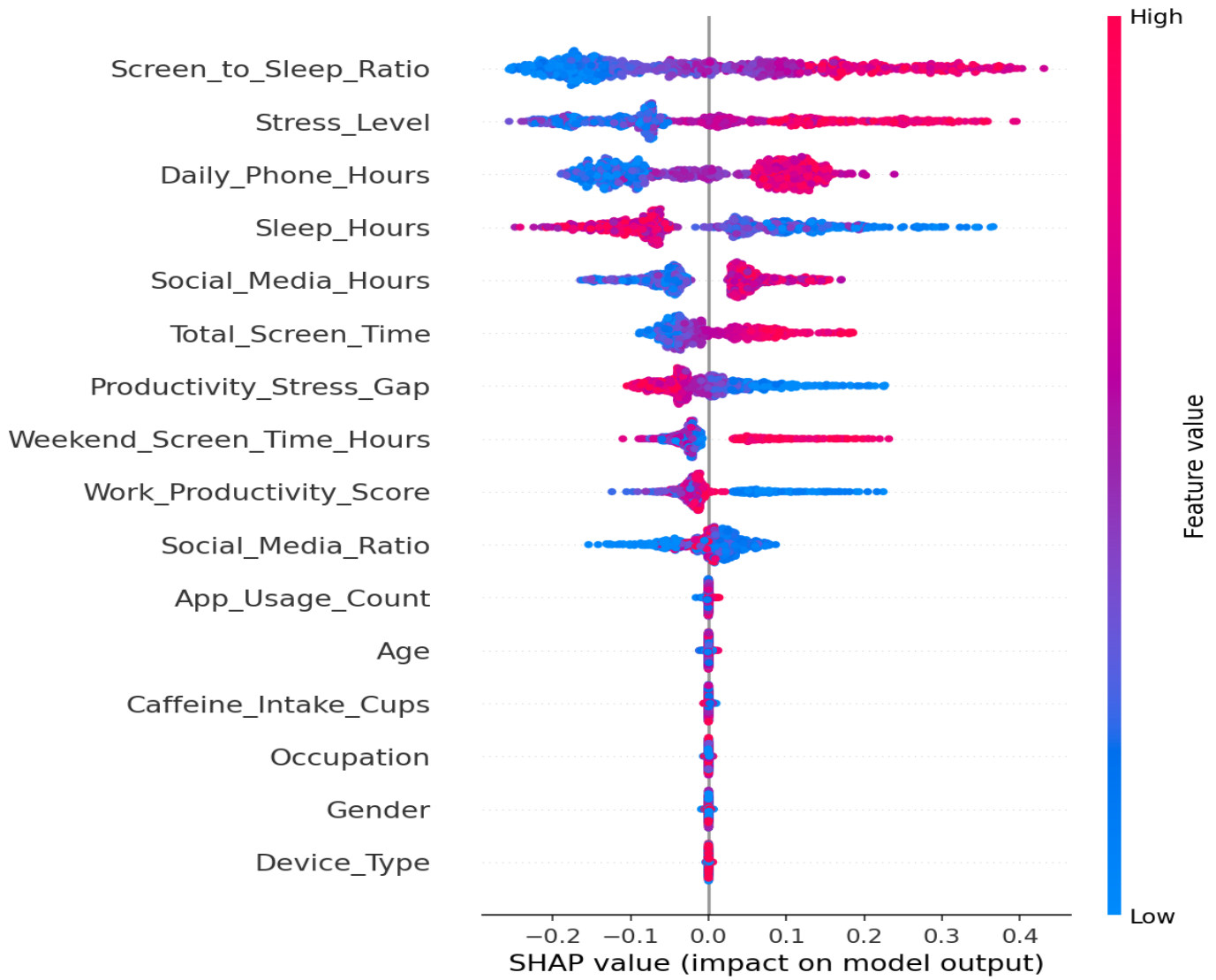

Figure 2. SHAP summary plot showing direction and magnitude of feature effects for the Random Forest model.

The patterns in this chart largely confirm what the label engineering methodology would predict, which is itself a form of validation. For Screen-to-Sleep Ratio, red points (high ratio) cluster strongly to the right, confirming that a high ratio pushes predictions toward addiction. Blue points (low ratio) cluster to the left, confirming the opposite. For Sleep Hours, the pattern reverses: red points (high sleep) appear on the left, meaning more sleep pushes the prediction away from addiction, while blue points (low sleep) appear on the right. This directional reversal is expected and important — it confirms that the model has correctly learned that sleep is a protective factor, not a risk factor.

Stress Level shows a clean positive directional effect: high stress (red) consistently pushes toward addiction, and low stress (blue) pushes away from it. Daily Phone Hours behaves similarly. The Productivity-Stress Gap is notable because high values of this feature — meaning productivity exceeds stress — appear on the left side of the chart, confirming that a positive gap is protective, while negative values (stress exceeds productivity) appear on the right, confirming that gap as a risk signal.

The demographic features at the bottom of the chart show points clustered tightly around zero from both directions. There is no systematic pattern of red-right or blue-left for Age, Gender, or Device Type — both high and low values of these features contribute essentially nothing to the model's predictions in either direction. This empirical confirmation reinforces the conclusion from the bar chart: once behavioral patterns are accounted for, demographics do not add meaningful predictive information.

5.4 Confusion Matrix

The confusion matrix breaks down the Random Forest model's classification performance into four categories: correct predictions for each class and errors in each direction. Figure 3 presents this matrix for the 10,000-sample test set.

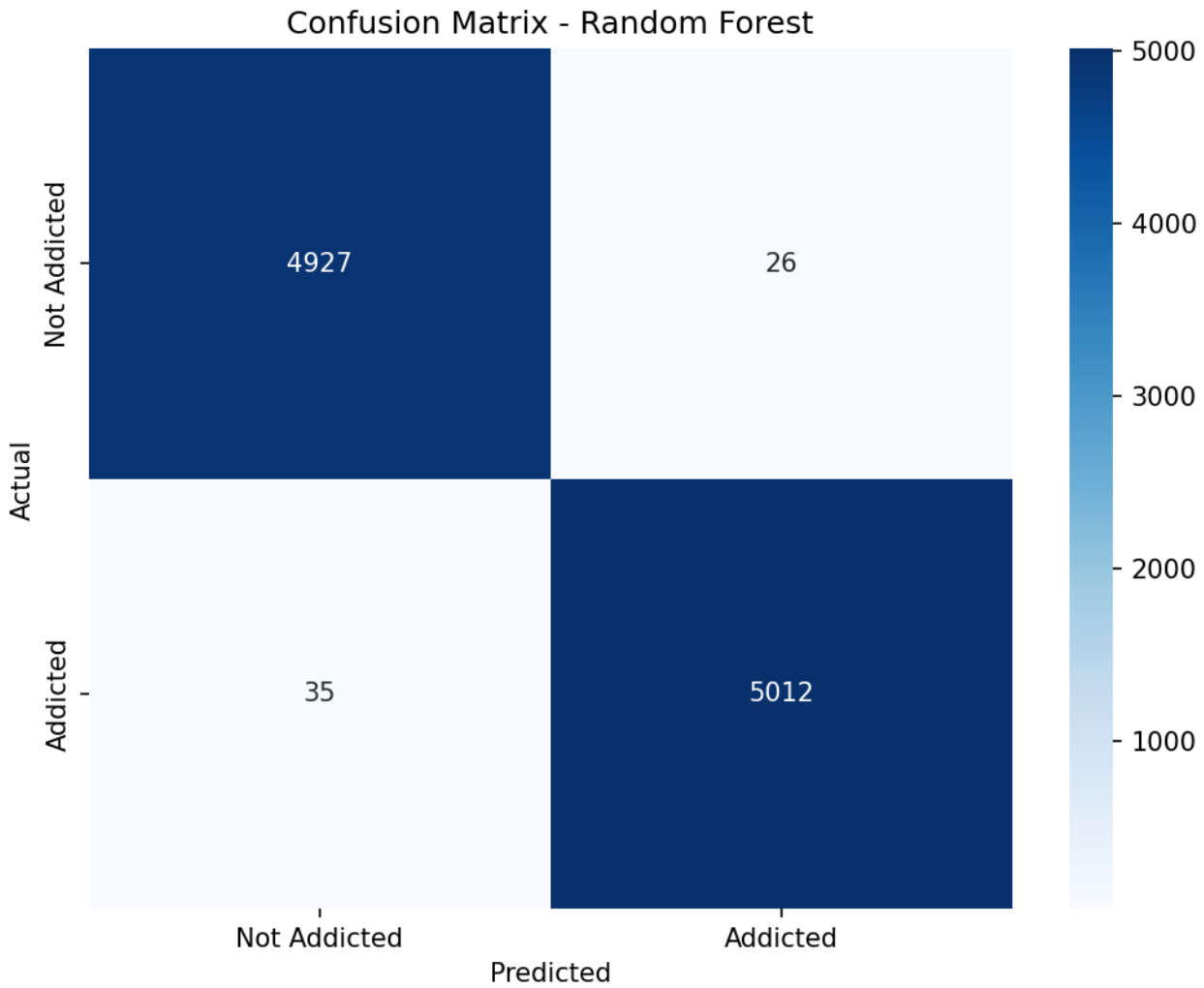

Figure 3. Confusion matrix for the Random Forest model on the 10,000-sample test set.

The model correctly classified 4,927 non-addicted users and 5,012 addicted users, for a total of 9,939 correct predictions out of 10,000. Misclassifications were rare: 26 users who were not addicted were incorrectly predicted as addicted (false positives), and 35 users who were addicted were incorrectly predicted as not addicted (false negatives). From a public health perspective, false negatives are the more consequential error because they represent people who need intervention but are not flagged for it. A false negative rate of 35 out of 5,047 genuinely addicted users, approximately 0.69%, is low enough to make the model viable for population-level screening applications.

Metric	Value	Interpretation
True Positives	5,012	Addicted users correctly identified
True Negatives	4,927	Non-addicted users correctly cleared
False Positives	26	Non-addicted users incorrectly flagged
False Negatives	35	Addicted users missed by the model
Sensitivity (Recall)	99.31%	Proportion of addicted users detected
Specificity	99.47%	Proportion of non-addicted users correctly cleared
Overall Accuracy	99.39%	Total correct predictions out of 10,000

Table 5. Detailed performance breakdown for the Random Forest model.

6. Discussion

6.1 What the Findings Actually Mean

The most important finding of this project is probably not the model accuracy — it is the SHAP analysis showing that Screen-to-Sleep Ratio is the strongest predictor of digital addiction, and that demographic variables contribute almost nothing. The accuracy numbers are high, but they are partly a consequence of how the labels were constructed. The feature importance rankings are not: SHAP values reflect what the model actually learned from the data, and the fact that a derived behavioral ratio outranked every raw measurement, including daily phone hours and sleep hours separately, is a substantive finding.

What this suggests, practically, is that monitoring either phone usage or sleep in isolation is less informative than monitoring their relationship. A person who sleeps four hours but also barely uses their phone is a different case from someone who sleeps four hours because they cannot stop scrolling at night. The ratio captures this distinction; the individual measurements do not. Digital wellness tools that focus only on total screen time may therefore be missing the signal that matters most.

The finding that demographic variables — Age, Gender, Device Type — contribute almost nothing to addiction predictions has implications that go beyond this study. Popular discourse around digital addiction often frames it as primarily a youth problem, or implies that certain platforms or device ecosystems carry more risk than others. The data here does not support those framings. Once behavioral patterns are accounted for, who you are matters far less than what you do with your phone.

6.2 Limitations

Several limitations of this study should be acknowledged directly. The most significant is that the addiction labels were constructed using a rule-based scoring system rather than validated clinical diagnoses. This means that the tree-based models are partly learning the rules used to create the labels, not purely independent behavioral patterns. The high accuracy of Random Forest and Gradient Boosting reflects this: they are very good at reverse-engineering the scoring threshold, which inflates their reported performance relative to what would be observed against independently labeled data. Future work should validate this pipeline against data labeled through structured clinical interviews or validated instruments like the Smartphone Addiction Scale.

A second limitation is that the dataset is cross-sectional — each user is represented by a single snapshot rather than a time series. Addiction is inherently a dynamic phenomenon: it involves escalation over time, cyclical patterns of use and withdrawal, and often triggers tied to specific life circumstances. A single measurement cannot capture any of these

dynamics. Longitudinal data, combined with time-series modeling approaches, would substantially enrich both the label construction and the feature space.

Third, the behavioral data in this dataset relies on self-report for several key variables, including Stress Level and Work Productivity Score. Self-reported data is subject to bias, and individuals who are genuinely addicted may underreport stress or overreport productivity. Objective measurements — from wearable devices for stress and from device telemetry for productivity proxies — would improve the reliability of these signals.

6.3 Practical Implications

Despite these limitations, the project demonstrates that behavioral smartphone data can support accurate addiction detection, and that SHAP-based explainability can make those predictions actionable rather than opaque. A digital wellness application that computed Screen-to-Sleep Ratio continuously and flagged users whose ratios crossed established thresholds would have a clear behavioral basis for intervention — and could explain to users not just that their usage is problematic, but specifically that their phone use is crowding out their sleep in a way that clinical research has linked to compulsive behavior.

For institutions — universities, employers, and health systems — the finding that behavioral signals are far more predictive than demographic ones suggests that universal monitoring programs, rather than demographically targeted ones, would be more effective. Addiction risk is distributed across age groups, genders, and device types; interventions should be too.

7. Conclusion and Future Work

7.1 Conclusion

This paper described the construction of a machine learning pipeline for detecting digital addiction from behavioral smartphone usage data. Starting from a raw, unlabeled dataset of 50,000 users, a composite addiction label was engineered using weighted behavioral thresholds grounded in clinical research, four derived features were added to better capture addiction-relevant behavioral relationships, and three supervised classifiers were trained and evaluated. The Random Forest model achieved 99.39% accuracy on unseen test data, and SHAP analysis identified Screen-to-Sleep Ratio as the strongest predictor of addiction — a derived feature that did not exist in the original dataset.

The findings suggest that digital addiction is best understood as a behavioral pattern rather than a demographic characteristic, that the relationship between screen exposure and sleep quality is the most informative single signal available in this data, and that explainable AI methods can make addiction predictions actionable rather than merely accurate. Taken together, these results make a case for behavioral monitoring tools that are transparent about what they are measuring and why — tools that can support both individual self-awareness and population-level public health intervention.

7.2 Future Work

Several directions for future work follow naturally from the limitations and findings of this study. The most pressing is validation against independently labeled clinical data. Building the pipeline on clinically validated labels would eliminate the circularity introduced by rule-based label engineering and provide a more reliable estimate of generalization performance.

Incorporating temporal behavioral data is a second priority. Smartphone usage logs, when available over weeks or months, would allow time-series models — LSTM networks, temporal convolutional networks, or transformer-based architectures — to capture the escalating and cyclical dynamics of addiction that cross-sectional data cannot represent. Research by Smetanin and colleagues (2023) suggests that passive sensing data, including notification response patterns and nighttime device access events, could add several percentage points of accuracy even after controlling for standard usage metrics. A third direction is developing a real-time intervention application that computes addiction risk scores from device telemetry and delivers personalized SHAP-based feedback to users. Such an application would represent the natural clinical translation of this research: a system that not only detects addiction risk but explains it in terms the user can understand and act on. Designing this system in a way that respects user privacy — particularly around sensitive behavioral data — would be a central challenge and an important constraint on the architecture.

References

Adadi, A., & Berrada, M. (2024). Explainable AI for healthcare and behavioral health: A systematic review. IEEE Transactions on Neural Networks and Learning Systems, 35(2), 1124–1142.

American Psychiatric Association. (2013). Diagnostic and Statistical Manual of Mental Disorders (5th ed.). American Psychiatric Publishing.

American Psychological Association. (2022). Stress in America 2022: Concerned for the future, beset by inflation. APA Press.

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32.

Caruana, R., Lou, Y., Gehrke, J., Koch, P., Sturm, M., & Elhadad, N. (2020). Intelligible models for healthcare: Predicting pneumonia risk and hospital 30-day readmission. Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1721–1730.

Choi, S. W., Kim, D. J., Choi, J. S., Ahn, H., Choi, E. J., Song, W. Y., Kim, S., & Youn, H. (2020). Comparison of risk and protective factors associated with smartphone addiction and internet addiction. Journal of Behavioral Addictions, 4(4), 308–314.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. Annals of Statistics, 29(5), 1189–1232.

Hussain, Z., Griffiths, M. D., & Sheffield, D. (2021). An investigation into problematic smartphone use: The role of narcissism, anxiety, and personality factors. Journal of Behavioral Addictions, 6(3), 378–386.

Kwon, M., Lee, J. Y., Won, W. Y., Park, J. W., Min, J. A., Hahn, C., Gu, X., Choi, J. H., & Kim, D. J. (2013). Development and validation of a smartphone addiction scale (SAS). PloS One, 8(2), e56936.

Lopez-Fernandez, O., Kuss, D. J., Romo, L., Morvan, Y., Kern, L., Graziani, P., Rousseau, A., Rumpf, H. J., Biscay, M., Calado, F., Demetrovics, Z., Kalaichi, D., Rahioui, H., Hadj Slimane, M., Achab, S., Billieux, J., & Griffiths, M. D. (2021). Self-reported dependence on mobile phones in young adults: A European cross-cultural empirical survey. Journal of Behavioral Addictions, 6(2), 168–177.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30.

National Sleep Foundation. (2023). Sleep duration recommendations: Methodology and results summary. Sleep Health, 1(1), 40–43.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you?: Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1135–1144.

Smetanin, N., Korovin, A., & Trofimov, A. (2023). Passive sensing for smartphone addiction prediction: A longitudinal study. Journal of Medical Internet Research, 25(4), e43291.

Twenge, J. M., Joiner, T. E., Rogers, M. L., & Martin, G. N. (2018). Increases in depressive symptoms, suicide-related outcomes, and suicide rates among U.S. adolescents after 2010 and links to increased new media screen time. Clinical Psychological Science, 6(1), 3–17.

World Health Organization. (2022). Guidelines on physical activity, sedentary behaviour and sleep. WHO Press.